Matthew Bernstein

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M.S. Data Science graduate with experience developing risk models, forecasting systems, and analytics dashboards in finance-oriented environments. Proficient in Python, SQL, R, and ETL, with a demonstrated ability to design and implement end-to-end machine learning pipelines. Seeking data science roles focused on predictive modeling and data-driven business decision systems.

Skills

Languages: Python, R, SQL
ML / Statistics: scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, regression, classification, time series, model evaluation
Data / Tools: Pandas, NumPy, Matplotlib, FastAPI, GitHub, GCP, Power BI, Tableau, Excel, Jira

Relevant Technical Work

Credit Default Risk API: GitHub

- Built a leakage-controlled LendingClub default-risk pipeline using 1.35M resolved loans, comparing four classification models. An isotonic-calibrated histogram gradient boosting model was selected using appropriate evaluation metrics.
- Deployed the model through FastAPI with individual and batch scoring, a static underwriting demo, Docker support, and automated tests; achieved 0.7092 test ROC-AUC and a 0.1463 Brier score, outperforming LendingClub grade and sub-grade baselines.

Bittbridge Miner-Validator Pipeline: GitHub

- Built live forecasting infrastructure for a student ML competition, ingesting 5-minute electricity load and weather data, storing features in Supabase, and enabling validators to score miner predictions.
- Built scraper and backend API workflows for robust subnet forecasting; created and tested finalized model pipelines including regression, decision trees, RNNs, and LSTMs.

Professional Experience

Bittbridge, UConn × Yuma — Storrs, CT
AI/ML Data Engineer Intern — *Spring 2026*

- Led technical design for a decentralized AI subnet project, defining pipeline architecture, miner submission workflows, validator scoring logic, and model evaluation criteria for live ML forecasting.
- Designed Supabase database schemas and data flow requirements for a miner–validator system, supporting feature storage, prediction retrieval, scoring, and repeatable experimentation across 80+ graduate students and AI practitioners.
- Built and presented a live GCP-based hackathon demo to 40+ data science and AI professionals, integrating blockchain, cloud, database, and ML components into a reproducible end-to-end forecasting environment.

Fidelity Investments — Jersey City, NJ
Data Analytics Intern - Tax — *Summer 2025*

- Analyzed 2M+ rows of financial transaction, sentiment, and web traffic datasets using SQL and ETL workflows to identify operational bottlenecks in frontend tax reporting processes.
- Built automated Power BI dashboards that track processing volume, associate workload, SLA trends, and exception patterns to reduce deliverables for tax operations managers.
- Streamlined Excel, SharePoint, and Jira workflows to improve task visibility and communication across for over 50 tax representatives.

Data Operations Intern - Tax — *Summer 2024*

- Integrated and transformed 100k+ rows of tax operations data using Excel Power Query workflows & SQL, supporting downstream reporting of tax compliance for US nonresidents.
- Performed ETL pipeline validation and audit testing with RPA and data engineering teams, providing support and ensuring structural integrity of production systems.

Education

University of Connecticut: The Graduate School — *Aug 2025 – July 2026*
M.S. in Data Science, Summer Capstone — GPA: 3.7/4.0

- **Relevant Coursework:** Machine Learning, Advanced Deep Learning, Applied Time Series, Applied Statistics, Data Mining

University of Connecticut — *Aug 2021 – May 2025*
B.A. in Applied Mathematical Sciences, Analytics Minor — GPA: 3.9/4.0

- **Relevant Coursework:** Mathematical Statistics I & II, Elementary Stochastic Processes, Database Design & Management (SQL), Business Information Systems (Excel), Data Visualization (Tableau)